

5.8

a) $(0.745 \text{ atm}) \left(\frac{760 \text{ mmHg}}{1 \text{ atm}} \right) = 566.2 = \mathbf{566 \text{ mmHg}}$

b) $(992 \text{ torr}) \left(\frac{1.01325 \text{ bar}}{760 \text{ torr}} \right) = 1.32256 = \mathbf{1.32 \text{ bar}}$

c) $(365 \text{ kPa}) \left(\frac{1 \text{ atm}}{101.325 \text{ kPa}} \right) = 3.60227 = \mathbf{3.60 \text{ atm}}$

d) $(804 \text{ mmHg}) \left(\frac{101.325 \text{ kPa}}{760 \text{ mmHg}} \right) = 107.191 = \mathbf{107 \text{ kPa}}$

5.18 At constant temperature and volume, the pressure of the gas is directly proportional to the number of moles of the gas. Verify this by examining the ideal gas equation. At constant T and V, the ideal gas equation becomes $P = n(RT/V)$ or $P = n \times \text{constant}$.

- 5.20
- a) As the pressure on a gas increases, the molecules move closer together, decreasing the volume. When the pressure is tripled, the **volume decreases to one third of the original volume** at constant temperature (Boyle's Law).
- b) As the temperature of a gas increases, the gas molecules gain kinetic energy. With higher energy, the gas molecules collide with the walls of the container with greater force, which increases the size (volume) of the container. If the temperature is increased by a factor of 2.5 (at constant pressure) then the **volume will increase by a factor of 2.5** (Charles's Law).
- c) As the number of molecules of gas increase, the force they exert on the container increases. This results in an increase in the volume of the container. Adding two moles of gas to one mole increases the number of moles by a factor of three, thus the **volume increases by a factor of three** (Avogadro's Law).

5.24 Plan: The temperature must be lowered to reduce the volume of a gas. Charles's Law states that at constant pressure and with a fixed amount of gas, the volume of a gas is directly proportional to the absolute temperature of the gas.

Solution:

$$(V_1 / T_1) = (V_2 / T_2) \quad \text{at constant } n \text{ and } P$$

$$T_2 = T_1 (V_2 / V_1)$$

$$T_2 = ((273 + 198) \text{ K}) \left(\frac{2.50 \text{ L}}{5.10 \text{ L}} \right) - 273 = -42.1167 = \mathbf{-42^\circ \text{C}}$$

5.26 Plan: To find V_2 , summarize and convert gas variables and apply the combined gas law equation.

$$(P_1 V_1 / T_1 = P_2 V_2 / T_2)$$

Solution:

$$V_2 = V_1 (T_2 / T_1) (P_1 / P_2) = (25.5 \text{ L}) \left(\frac{273 \text{ K}}{298 \text{ K}} \right) \left(\frac{153.3 \text{ kPa}}{101.325 \text{ kPa}} \right) = 35.3437 = \mathbf{35.3 \text{ L}}$$

5.28 Plan: Given the volume, pressure, and temperature of a gas, the number of moles of the gas can be calculated using the ideal gas equation, $n = PV/RT$. The gas constant, $R = 0.0821 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$, gives pressure in atmospheres and temperature in Kelvin. The given pressure in torr must be converted to atmospheres and the temperature converted to Kelvin.

Solution:

$$n = PV / RT = \frac{(228 \text{ torr})(5.0 \text{ L})}{\left(0.0821 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}\right)((273+27) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}} \right) = 0.060901 = \mathbf{0.061 \text{ mol chlorine}}$$

5.30 Plan: Solve the ideal gas equation for moles and convert to mass using the molar mass of ClF_3 .

Solution:

$$PV = (m / \mathcal{M})RT$$

$$m = PV \mathcal{M} / RT = \frac{(699 \text{ mmHg})(207 \text{ mL}) \left(92.45 \frac{\text{g}}{\text{mol}} \right)}{\left(0.0821 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \right)((273 + 45) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ mmHg}} \right) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \\ = 0.67417 = \mathbf{0.674 \text{ g ClF}_3}$$

5.33 Plan: Assuming that while rising in the atmosphere the balloon will neither gain nor lose gas molecules, the number of moles of gas calculated at sea level will be the same as the number of moles of gas at the higher altitude. Using the ideal gas equation, $(n \times R)$ is a constant equal to (PV/T) . Given the sea-level conditions of volume, pressure and temperature, and the temperature and pressure at the higher altitude for the gas in the balloon, we can set up an equation to solve for the volume at the higher altitude. Comparing the calculated volume to the given maximum volume of 835 L will tell us if the balloon has reached its maximum volume at this altitude.

Solution:

$$(P_1 V_1 / T_1) = (P_2 V_2 / T_2)$$

$$V_2 = (P_1 V_1 T_2 / T_1 P_2) = \frac{(755 \text{ torr})(55.0 \text{ L})((273 - 5) \text{ K})}{((273 + 23) \text{ K})(0.066 \text{ atm})} \left(\frac{1 \text{ atm}}{760 \text{ torr}} \right) = 749.54 = 7.5 \times 10^2 \text{ L}$$

The calculated volume of the gas at the higher altitude is less than the maximum volume of the balloon. **No**, the balloon will not reach its maximum volume.

Check: Should we expect that the volume of the gas in the balloon should increase? At the higher altitude, the pressure decreases, this increases the volume of the gas. At the higher altitude, the temperature decreases, this decreases the volume of the gas. Which of these will dominate? The pressure decreases by a factor of $0.99/0.066 = 15$. If we label the initial volume V_1 , then the resulting volume is $15 V_1$. The temperature decreases by a factor of $296/268 = 1.1$, so the resulting volume is $V_1/1.1$ or $0.91 V_1$. The increase in volume due to the change in pressure is greater than the decrease in volume due to change in temperature, so the volume of gas at the higher altitude should be greater than the volume at sea level.

5.35 The molar mass of H_2 is less than the average molar mass of air (mostly N_2 , O_2 , and Ar), so air is denser. To collect a beaker of $\text{H}_2(g)$, **invert** the beaker so that the air will be replaced by the lighter H_2 . The molar mass of CO_2 is greater than the average molar mass of air, so $\text{CO}_2(g)$ is more dense. Collect the CO_2 holding the beaker **upright**, so the lighter air will be displaced out the top of the beaker.

5.38 Plan: Using the ideal gas equation and the molar mass of xenon, 131.3 g/mol, we can find the density of xenon gas at STP. Standard temperature is 0°C and standard pressure is 1 atm. Do not forget that the pressure at STP is exact and will not affect the significant figures.

Solution:

$$d = \mathcal{M} P / RT = \frac{(131.3 \text{ g/mol})(1 \text{ atm})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(273 \text{ K})} = 5.8581 = \mathbf{5.86 \text{ g/L}}$$

5.40 Plan: Apply the ideal gas equation to determine the number of moles. Convert moles to mass and divide by the volume to obtain density in g/L. Do not forget that the pressure at STP is exact and will not affect the significant figures.

Solution:

$$n = PV / RT = \frac{(1 \text{ atm})(0.0400 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(273 \text{ K})} = 1.78465 \times 10^{-3} = \mathbf{1.78 \times 10^{-3} \text{ mol AsH}_3}$$

$$d = \text{mass} / \text{volume} = \frac{(1.78465 \times 10^{-3} \text{ mol})(77.94 \text{ g/mol})}{(0.0400 \text{ L})} = 3.47740 = \mathbf{3.48 \text{ g/L}}$$

5.42 Plan: Rearrange the formula $PV = (m / \mathcal{M})RT$ to solve for molar mass: $\mathcal{M} = mRT / PV$.

Solution:

$$\begin{aligned} \mathcal{M} = mRT / PV &= \frac{(206 \text{ ng})\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 45)\text{K})}{(388 \text{ torr})(0.206 \mu\text{L})} \left(\frac{10^{-9} \text{ g}}{1 \text{ ng}}\right) \left(\frac{1 \mu\text{L}}{10^{-6} \text{ L}}\right) \left(\frac{760 \text{ torr}}{1 \text{ atm}}\right) \\ &= 51.13899 = \mathbf{51.1 \text{ g/mol}} \end{aligned}$$

- 5.44 Plan: Use the ideal gas equation to determine the number of moles of Ar and O₂. The gases are combined ($n_{\text{TOT}} = n_{\text{Ar}} + n_{\text{O}}$) into a 400 mL flask (V) at 27°C (T). Determine the total pressure from n_{TOT} , V, and T.

Solution:

$$n = PV / RT$$

$$\text{Moles Ar} = PV / RT = \frac{(1.20 \text{ atm})(0.600 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 227) \text{ K})} = 0.017539585 \text{ mol Ar (unrounded)}$$

$$\text{Moles O}_2 = PV / RT = \frac{(501 \text{ torr})(0.200 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 127) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) = 0.004014680 \text{ mol O}_2 \text{ (unrounded)}$$

$$\begin{aligned} P_{\text{mixture}} = nRT / V &= \frac{((0.017539585 + 0.004014680) \text{ mol}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 27) \text{ K})}{400 \text{ mL}} \left(\frac{1 \text{ mL}}{10^{-3} \text{ L}}\right) \\ &= 1.32720 = \mathbf{1.33 \text{ atm}} \end{aligned}$$

- 5.48 Plan: The problem gives the mass, volume, temperature and pressure of a gas, so we can solve for molar mass using $\mathcal{M} = mRT/PV$. The problem also states that the gas is a hydrocarbon, which by, definition, contains only carbon and hydrogen atoms. We are also told that each molecule of the gas contains five carbon atoms so we can use this information and the calculated molar mass to find out how many hydrogen atoms are present and the formula of the compound.

Solution:

$$\mathcal{M} = mRT/PV = \frac{(0.482 \text{ g}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 101) \text{ K})}{(767 \text{ torr})(0.204 \text{ L})} \left(\frac{760 \text{ torr}}{1 \text{ atm}}\right) = 71.8869 \text{ g/mol (unrounded)}$$

The carbon accounts for $[5 (12 \text{ g/mol})] = 60 \text{ g/mol}$, thus, the hydrogen must make up the difference $(72 - 60) = 12 \text{ g/mol}$. A value of 12 g/mol corresponds to 12 H atoms. (Since fractional atoms are not possible, rounding is acceptable.)

Therefore, the molecular formula is **C₅H₁₂**.

- 5.50 a) $PV = nRT$

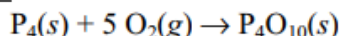
$$n = PV / RT = \frac{(850. \text{ torr})(21 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 45) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) = 0.89961 = \mathbf{0.90 \text{ mol gas}}$$

b) The information given in ppm is a way of expressing the proportion, or fraction, of SO₂ present in the mixture. Since n is directly proportional to V , the *volume* fraction can be used in place of the *mole* fraction used in equation 5.12. There are 7.95×10^3 parts SO₂ in a million parts of mixture, so volume fraction = $(7.95 \times 10^3 / 1 \times 10^6) = 7.95 \times 10^{-3}$.

Therefore, $P_{\text{SO}_2} = \text{volume fraction} \times P_{\text{TOT}} = (7.95 \times 10^{-3})(850. \text{ torr}) = 6.7575 = \mathbf{6.76 \text{ torr}}$.

- 5.51 Plan: We can find the moles of oxygen from the standard molar volume of gases and use the stoichiometric ratio from the balanced equation to determine the moles of phosphorus that will react with the oxygen.

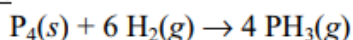
Solution:



$$\text{Mass P}_4 = (35.5 \text{ L O}_2) \left(\frac{1 \text{ mol O}_2}{22.4 \text{ L O}_2}\right) \left(\frac{1 \text{ mol P}_4}{5 \text{ mol O}_2}\right) \left(\frac{123.88 \text{ g P}_4}{1 \text{ mol P}_4}\right) = 39.2655 = \mathbf{39.3 \text{ g P}_4}$$

- 5.53 Plan: To find the mass of PH_3 , write the balanced equation, convert mass of P_4 to moles, solve for moles of H_2 using the standard molar volume (or use ideal gas equation), and proceed with the limiting reactant problem.

Solution:



Moles hydrogen = $(83.0 \text{ L}) (1 \text{ mol} / 22.4 \text{ L}) = 3.705357 \text{ mol H}_2$ (unrounded)

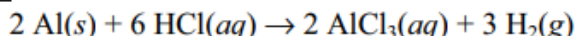
Moles phosphorus = $(37.5 \text{ g P}_4) (1 \text{ mol P}_4 / 123.88 \text{ g P}_4) = 0.302712 \text{ mol P}_4$ (unrounded)

Dividing each of the moles by the coefficients shows that P_4 is the limiting reactant.

$$\text{Mass PH}_3 = (37.5 \text{ g P}_4) \left(\frac{1 \text{ mol P}_4}{123.88 \text{ g P}_4} \right) \left(\frac{4 \text{ mol PH}_3}{1 \text{ mol P}_4} \right) \left(\frac{33.99 \text{ g PH}_3}{1 \text{ mol PH}_3} \right) = 41.15676 = \mathbf{41.2 \text{ g PH}_3}$$

- 5.55 Plan: First, write the balanced equation. The moles of hydrogen produced can be calculated from the ideal gas equation and then the stoichiometric ratio from the balanced equation is used to determine the moles of aluminum that reacted. The problem specifies “hydrogen gas collected over water,” so the partial pressure of water must first be subtracted. Table 5.3 reports pressure at 26°C (25.2 torr) and 28°C (28.3 torr), so take the average of the two values.

Solution:



Hydrogen pressure = total pressure - pressure from water =

$$(751 \text{ mmHg}) - [(28.3 + 25.2) \text{ torr} / 2] = 724.25 \text{ torr (unrounded)}$$

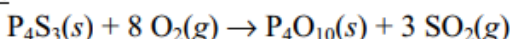
Moles of hydrogen = $n = PV / RT$

Mass of aluminum = mol of hydrogen x mole ratio x molar mass of Al = PV/RT x mole ratio x molar mass of Al

$$\begin{aligned} \text{Mass of Al} &= \left[\frac{(724.25 \text{ torr})(35.8 \text{ mL})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 27) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}} \right) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \right] \left(\frac{2 \text{ mol Al}}{3 \text{ mol H}_2} \right) \left(\frac{26.98 \text{ g Al}}{1 \text{ mol Al}} \right) \\ &= 0.024914 = \mathbf{0.0249 \text{ g Al}} \end{aligned}$$

- 5.57 Plan: To find mL of SO_2 , write the balanced equation, convert the given mass of P_4S_3 to moles, use the molar ratio to find moles of SO_2 , and use the ideal gas equation to find volume.

Solution:



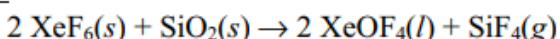
Moles SO_2 = mass P_4S_3 x 1/molar mass P_4S_3 x mole ratio

Volume SO_2 = nRT / P = (mass P_4S_3 x 1/molar mass P_4S_3 x mole ratio) RT / P

$$\begin{aligned} V &= \frac{\left[(0.800 \text{ g P}_4\text{S}_3) \left(\frac{1 \text{ mol P}_4\text{S}_3}{220.09 \text{ g P}_4\text{S}_3} \right) \left(\frac{3 \text{ mol SO}_2}{1 \text{ mol P}_4\text{S}_3} \right) \right] \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) ((273 + 32) \text{ K})}{725 \text{ torr}} \left(\frac{760 \text{ torr}}{1 \text{ atm}} \right) \left(\frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) \\ &= 286.239 = \mathbf{286 \text{ mL}} \end{aligned}$$

- 5.59 Plan: First write the balanced equation. Given the amount of xenon hexafluoride that reacts, we can find the number of moles of silicon tetrafluoride gas formed. Then, using the ideal gas equation with the moles of gas, the temperature and the volume, we can calculate the pressure of the silicon tetrafluoride gas.

Solution:



Mole SiF_4 = n = mass XeF_6 x 1/molar mass XeF_6 x mole ratio

Pressure SiF_4 = $P = nRT / V$ = (mass XeF_6 x 1/molar mass XeF_6 x mole ratio) RT / V

$$\begin{aligned} P &= \frac{\left[(2.00 \text{ g XeF}_6) \left(\frac{1 \text{ mol XeF}_6}{245.3 \text{ g XeF}_6} \right) \left(\frac{1 \text{ mol SiF}_4}{2 \text{ mol XeF}_6} \right) \right] \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) ((273 + 25) \text{ K})}{1.00 \text{ L}} \\ &= 0.099738 = \mathbf{0.0997 \text{ atm SiF}_4} \end{aligned}$$

5.63 At STP (or any identical temperature and pressure), the volume occupied by a mole of any gas will be identical. This is because at the same temperature, all gases have the same average kinetic energy, resulting in the same pressure.

5.66 The molar masses of the three gases are 2.016 for H_2 (Flask A), 4.003 for He (Flask B), and 16.04 for CH_4 (Flask C). Since hydrogen has the smallest molar mass of the three gases, 4 g of H_2 will contain more gas molecules than 4 g of He or 4 g of CH_4 . Since helium has a smaller molar mass than methane, 4 g of He will contain more gas molecules than 4 g of CH_4 .

a) $P_A > P_B > P_C$ The pressure of a gas is proportional to the number of gas molecules. So, the gas sample with more gas molecules will have a greater pressure.

b) $E_A = E_B = E_C$ Average kinetic energy depends only on temperature. The temperature of each gas sample is 273 K, so they all have the same average kinetic energy.

c) $\text{rate}_A > \text{rate}_B > \text{rate}_C$ When comparing the speed of two gas molecules, the one with the lower mass travels faster.

d) $\text{total } E_A > \text{total } E_B > \text{total } E_C$ Since the average kinetic energy for each gas is the same (part b of this problem) then the total kinetic energy would equal the average times the number of molecules. Since the hydrogen flask contains the most molecules, its total kinetic energy will be the greatest.

e) $d_A = d_B = d_C$ Under the conditions stated in this problem, each sample has the same volume, 5 L, and the same mass, 4 g. Thus, the density of each is $4 \text{ g}/5 \text{ L} = 0.8 \text{ g/L}$.

f) **Collision frequency (A) > collision frequency (B) > collision frequency (C)** The number of collisions depends on both the speed and the distance between gas molecules. Since hydrogen is the lightest molecule it has the greatest speed and the 5 L flask of hydrogen also contains the most molecules, so collisions will occur more frequently between hydrogen molecules than between helium molecules. By the same reasoning, collisions will occur more frequently between helium molecules than between methane molecules.

5.67 To find the ratio of effusion rates, calculate the inverse of the ratio of the square roots of the molar masses (equation 5.14).

$$\frac{\text{Rate } \text{H}_2}{\text{Rate } \text{UF}_6} = \sqrt{\frac{\text{Molar Mass } \text{UF}_6}{\text{Molar Mass } \text{H}_2}} = \sqrt{\frac{352.0 \text{ g/mol}}{2.016 \text{ g/mol}}} = 13.2137 = \mathbf{13.21}$$

5.69 a) The gases have the same average kinetic energy because they are at the same temperature. The heavier Ar atoms are moving slower than the lighter He atoms to maintain the same average kinetic energy. Therefore, **Curve 1** better represents the behavior of Ar.

b) A gas that has a slower molecular speed would effuse more slowly, so **Curve 1** is the better choice.

c) Fluorine gas exists as a diatomic molecule, F_2 , with $\mathcal{M} = 38.00 \text{ g/mol}$. Therefore, F_2 is much closer in size to Ar than He, so **Curve 1** more closely represents F_2 's behavior.

$$5.71 \quad \frac{\text{Rate He}}{\text{Rate F}_2} = \frac{\text{Time F}_2}{\text{Time He}} = \sqrt{\frac{\text{Molar Mass F}_2}{\text{Molar Mass He}}} = \sqrt{\frac{38.00 \text{ g/mol}}{4.003 \text{ g/mol}}} = 3.08105 \text{ (unrounded)}$$

$$\text{Time F}_2 = 3.08105 (\text{time He}) = 3.08105 (4.55 \text{ min}) = 14.018786 = \mathbf{14.0 \text{ min}}$$

- 5.73 White phosphorus is a molecular form of the element phosphorus consisting of some number, x, of phosphorus atoms. Determine the number of phosphorus atoms, x, in one molecule of white phosphorus from the rate of effusion of the gaseous phosphorus molecules.

$$\frac{\text{Rate P}_x}{\text{Rate Ne}} = \sqrt{\frac{\text{Molar Mass Ne}}{\text{Molar Mass P}_x}} = 0.404$$

$$\frac{\text{Molar Mass Ne}}{\text{Molar Mass P}_x} = (0.404)^2 = 0.163216 \text{ (unrounded)}$$

$$\text{Molar Mass P}_x = \text{Molar Mass Ne} / 0.163216 = 20.18 / 0.163216 = 123.6398 \text{ g/mol}$$

$$\text{Molar Mass P}_x = x (\text{Molar Mass P}) = x (30.97 \text{ g/mol})$$

$$x = (123.6398 \text{ g/mol}) / (30.97 \text{ g/mol}) = 3.9922 = 4$$

Thus, **4 atoms per molecule**, so $\text{P}_x = \text{P}_4$.

- 5.77 Nitrogen gas behaves more ideally at 1 atm than at 500 atm because at lower pressures the gas molecules are farther apart. An ideal gas is defined as consisting of gas molecules that act independently of the other gas molecules. When gas molecules are far apart, they act ideally, because intermolecular attractions are less important and the volume of the molecules is a smaller fraction of the container volume.
- 5.80 Molar mass has units of g/mol, so solve for the number of hemoglobin (Hb) moles combined with O₂. Moles of oxygen (from ideal gas equation) combine with Hb in a 4:1 ratio.
 $PV = nRT$

$$\text{Moles O}_2 = PV / RT = \frac{(743 \text{ torr})(1.53 \text{ mL})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 37) \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}}\right)$$

$$= 5.87708 \times 10^{-5} \text{ mol O}_2 \text{ (unrounded)}$$

$$\text{Moles Hb} = (5.87708 \times 10^{-5} \text{ mol O}_2) (1 \text{ mol Hb} / 4 \text{ mol O}_2) = 1.46927 \times 10^{-5} \text{ mol Hb (unrounded)}$$

$$\text{Molar mass hemoglobin} = (1.00 \text{ g Hb}) / (1.46927 \times 10^{-5} \text{ mol Hb}) = 6.806098 \times 10^4 = \mathbf{6.81 \times 10^4 \text{ g/mol}}$$

- 5.83 a) Ideal gas equation: $PV = nRT$

$$P_{\text{IGL}} = nRT/V = \frac{\left[(0.5850 \text{ kg Cl}_2) \left(\frac{10^3 \text{ g}}{1 \text{ kg}}\right) \left(\frac{1 \text{ mol Cl}_2}{70.90 \text{ g Cl}_2}\right)\right] \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) ((273 + 225) \text{ K})}{15.00 \text{ L}}$$

$$= 22.4791 = \mathbf{22.5 \text{ atm}}$$

b) $\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$

$$P_{\text{VDW}} = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

$$n = (0.5850 \text{ kg}) (10^3 \text{ g/1 kg}) (1 \text{ mol Cl}_2 / 70.90 \text{ g Cl}_2) = 8.251078 \text{ mol Cl}_2 \text{ (unrounded)}$$

$$P_{\text{VDW}} = \frac{(8.251078 \text{ mol Cl}_2) \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) ((273 + 225) \text{ K})}{15.00 \text{ L} - (8.251078 \text{ mol Cl}_2) \left(0.0562 \frac{\text{L}}{\text{mol}}\right)} - \frac{(8.251078 \text{ mol Cl}_2)^2 \left(6.49 \frac{\text{atm} \cdot \text{L}^2}{\text{mol}^2}\right)}{(15.00 \text{ L})^2}$$

$$= 21.23252 = \mathbf{21.2 \text{ atm}}$$

5.86 a) Partial pressures are calculated from Dalton's Law of Partial Pressures: $P_A = X_A(P_{\text{total}})$

Convert each mole percent to a mole fraction by dividing by 100%.

$$P_{\text{Nitrogen}} = X_{\text{Nitrogen}} P_{\text{Total}} = (0.786) (1.00 \text{ atm}) (760 \text{ torr} / 1 \text{ atm}) = 597.36 = \mathbf{597 \text{ torr } N_2}$$

$$P_{\text{Oxygen}} = X_{\text{Oxygen}} P_{\text{Total}} = (0.209) (1.00 \text{ atm}) (760 \text{ torr} / 1 \text{ atm}) = 158.84 = \mathbf{159 \text{ torr } O_2}$$

$$P_{\text{Carbon Dioxide}} = X_{\text{Carbon Dioxide}} P_{\text{Total}} = (0.0004) (1.00 \text{ atm}) (760 \text{ torr} / 1 \text{ atm}) = 0.304 = \mathbf{0.3 \text{ torr } CO_2}$$

$$P_{\text{Water}} = X_{\text{Water}} P_{\text{Total}} = (0.0046) (1.00 \text{ atm}) (760 \text{ torr} / 1 \text{ atm}) = 3.496 = \mathbf{3.5 \text{ torr } O_2}$$

b) Mole fractions can be calculated by rearranging Dalton's Law of Partial Pressures:

$X_A = P_A/P_{\text{total}}$ and multiply by 100 to express mole fraction as percent.

$$P_{\text{Total}} = (569 + 104 + 40 + 47) \text{ torr} = 760 \text{ torr}$$

$$N_2: [(569 \text{ torr}) / (760 \text{ torr})] \times 100\% = 74.7684 = \mathbf{74.9 \text{ mol}\% } N_2$$

$$O_2: [(104 \text{ torr}) / (760 \text{ torr})] \times 100\% = 13.6842 = \mathbf{13.7 \text{ mol}\% } O_2$$

$$CO_2: [(40 \text{ torr}) / (760 \text{ torr})] \times 100\% = 5.263 = \mathbf{5.3 \text{ mol}\% } CO_2$$

$$H_2O: [(47 \text{ torr}) / (760 \text{ torr})] \times 100\% = 6.1842 = \mathbf{6.2 \text{ mol}\% } CO_2$$

c) Number of molecules of O_2 can be calculated using the Ideal Gas Equation and Avogadro's number.

$$\begin{aligned} \text{Molecules } O_2 &= \frac{(104 \text{ torr})(0.50 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 37)\text{K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) \left(\frac{6.022 \times 10^{23} O_2 \text{ molecules}}{1 \text{ mol } O_2}\right) \\ &= 1.61892 \times 10^{21} = \mathbf{1.6 \times 10^{21} \text{ molecules } O_2} \end{aligned}$$

$$5.88 \quad a) V_f = P_i V_i T_f / T_i P_f = \frac{(1400. \text{ mmHg})(208 \text{ mL})(298 \text{ K})}{(286 \text{ K})(1 \text{ atm})} \left(\frac{1 \text{ atm}}{760 \text{ mmHg}} \right) = 399.23 = \mathbf{4 \times 10^2 \text{ mL}}$$

$$b) n = PV / RT = \frac{(1400. \text{ mmHg})(208 \text{ mL})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (286 \text{ K})} \left(\frac{77\% \text{ N}_2}{100\%} \right) \left(\frac{1 \text{ atm}}{760 \text{ mmHg}} \right) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right)$$

$$= 0.0125649 = \mathbf{0.013 \text{ mol N}_2}$$

- 5.95 The empirical formula for aluminum chloride is AlCl_3 ($\mathcal{M} = 133.33 \text{ g/mol}$). Calculate the molar mass of the gaseous species from the ratio of effusion rates. This molar mass, divided by the empirical weight, should give a whole number multiple that will yield the molecular formula.

$$\frac{\text{Rate Unk}}{\text{Rate He}} = 0.122 = \sqrt{\frac{\text{Molar Mass He}}{\text{Molar Mass Unk}}}$$

$$0.122 = \sqrt{\frac{4.003 \text{ g/mol}}{\text{Molar Mass Unk}}}$$

Molar Mass Unknown = 270.7605 g/mol (unrounded)

The whole number multiple is $270.7605/133.33$, which is about 2. Therefore, the molecular formula of the gaseous species is $2 \times (\text{AlCl}_3) = \mathbf{\text{Al}_2\text{Cl}_6}$.

- 5.97 Plan: First, write the balanced equation for the reaction: $2 \text{SO}_2 + \text{O}_2 \rightarrow 2 \text{SO}_3$. The total number of moles of gas will change as the reaction occurs since 3 moles of reactant gas forms 2 moles of product gas. From the volume, temperature and pressures given, we can calculate the number of moles of gas before and after the reaction. For each mole of SO_3 formed, the total number of moles of gas decreases by $1/2$ mole. Thus, twice the decrease in moles of gas equals the moles of SO_3 formed.

Solution:

Moles of gas before and after reaction

$$\text{Initial moles} = PV / RT = \frac{(1.95 \text{ atm})(2.00 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (900. \text{ K})} = 0.05278116 \text{ mol (unrounded)}$$

$$\text{Final moles} = PV / RT = \frac{(1.65 \text{ atm})(2.00 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (900. \text{ K})} = 0.04466098 \text{ mol (unrounded)}$$

Moles of SO_3 produced = $2 \times$ decrease in the total number of moles

$$= 2 \times (0.05278116 \text{ mol} - 0.04466098 \text{ mol})$$

$$= 0.01624035 = \mathbf{1.62 \times 10^{-2} \text{ mol}}$$

Check: If the starting amount is 0.0528 total moles of SO_2 and O_2 , then $x + y = 0.0528 \text{ mol}$, where $x = \text{mol of SO}_2$ and $y = \text{mol of O}_2$. After the reaction:

$$(x - z) + (y - 0.5z) + z = 0.0447 \text{ mol}$$

Where $z = \text{mol of SO}_3 \text{ formed} = \text{mol of SO}_2 \text{ reacted} = 2(\text{mol of O}_2 \text{ reacted})$.

Subtracting the two equations gives:

$$x - (x - z) + y - (y - 0.5z) - z = 0.0528 - 0.0447$$

$$z = 0.0163 \text{ mol SO}_3$$

The approach of setting up two equations and solving them gives the same result as above.

- 5.101 a) The balanced equation for the reaction is $\text{Ni(s)} + 4 \text{CO(g)} \rightarrow \text{Ni(CO)}_4\text{(g)}$. Although the problem states that Ni is impure, you can assume the impurity is unreactive and thus not include it in the reaction. The mass of Ni can be calculated from the stoichiometric relationship between moles of CO (using the ideal gas equation) and Ni.
 Mass Ni = mol Ni x molar mass

$$\text{Mass Ni} = \frac{(100.7 \text{ kPa})(3.55 \text{ m}^3)}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 50) \text{ K})} \left(\frac{1 \text{ atm}}{101.325 \text{ kPa}}\right) \left(\frac{1 \text{ L}}{10^{-3} \text{ m}^3}\right) \left(\frac{1 \text{ mol Ni}}{4 \text{ mol CO}}\right) \left(\frac{58.69 \text{ g Ni}}{1 \text{ mol Ni}}\right)$$

$$= 1952.089 = \mathbf{1.95 \times 10^3 \text{ g Ni}}$$

- b) Assume the volume is 1 m³. Use the ideal gas equation to solve for moles of Ni(CO)₄, which equals the moles of Ni, and convert moles to grams using the molar mass.

$$\text{Mass Ni} = \frac{(21 \text{ atm})(1 \text{ m}^3)}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)((273 + 155) \text{ K})} \left(\frac{1 \text{ L}}{10^{-3} \text{ m}^3}\right) \left(\frac{1 \text{ mol Ni}}{1 \text{ mol Ni(CO)}_4}\right) \left(\frac{58.69 \text{ g Ni}}{1 \text{ mol Ni}}\right)$$

$$= 3.50749 \times 10^4 = \mathbf{3.5 \times 10^4 \text{ g Ni}}$$

The pressure limits the significant figures.

- c) The amount of CO needed to form Ni(CO)₄ is the same amount that is released on decomposition. The vapor pressure of water at 35°C (42.2 torr) must be accounted for (see Table 5.3). Use the ideal gas equation to calculate the volume of CO, and compare this to the volume of Ni(CO)₄. The mass of Ni from a m³ (part b) can be used to calculate the amount of CO released.

$$P_{\text{CO}} = P_{\text{total}} - P_{\text{water}} = 769 \text{ torr} - 42.2 \text{ torr} = 726.8 \text{ torr (unrounded)}$$

$$V = nRT / P =$$

$$\frac{\left[3.50749 \times 10^4 \text{ g Ni} \left(\frac{1 \text{ mol Ni}}{58.69 \text{ g Ni}}\right) \left(\frac{4 \text{ mol CO}}{1 \text{ mol Ni}}\right)\right] \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) ((273 + 35) \text{ K})}{726.8 \text{ torr}} \left(\frac{760 \text{ torr}}{1 \text{ atm}}\right) \left(\frac{10^{-3} \text{ m}^3}{1 \text{ L}}\right)$$

$$= 63.209866 = \mathbf{63 \text{ m}^3 \text{ CO}}$$

The answer is limited to two significant figures because the mass of Ni comes from part b.

- 5.103 a) A preliminary equation for this reaction is $4 \text{C}_x\text{H}_y\text{N}_z + n \text{O}_2 \rightarrow 4 \text{CO}_2 + 2 \text{N}_2 + 10 \text{H}_2\text{O}$. Since the organic compound does not contain oxygen, the only source of oxygen as a reactant is oxygen gas. To form 4 volumes of CO₂ would require 4 volumes of O₂ and to form 10 volumes of H₂O would require 5 volumes of O₂. Thus, **9 volumes of O₂** was required.
- b) Since the volume of a gas is proportional to the number of moles of the gas we can equate volume and moles. From a volume ratio of 4 CO₂:2 N₂:10 H₂O we deduce a mole ratio of 4 C:4 N:20 H or 1 C:1 N:5 H for an empirical formula of **CH₅N**.

- 5.107 To find the factor by which a diver's lungs would expand, find the factor by which P changes from 125 ft to the surface, and apply Boyle's Law. To find that factor, calculate P_{seawater} at 125 ft by converting the given depth from ft-seawater to mmHg to atm and adding the surface pressure (1.00 atm).
 $P(\text{H}_2\text{O}) = (125 \text{ ft}) (12 \text{ in} / \text{ft}) (2.54 \text{ cm} / 1 \text{ in}) (10^{-2} \text{ m} / 1 \text{ cm}) (1 \text{ mm} / 10^{-3} \text{ m}) = 3.81 \times 10^4 \text{ mmHg}$
 $P(\text{Hg}) = (3.81 \times 10^4 \text{ mmHg}) [(1.04 \text{ g H}_2\text{O/mL}) / (13.5 \text{ g Hg/mL})] (1 \text{ atm} / 760 \text{ mmHg})$
 $= 3.861988 \text{ atm (unrounded)}$
 $P_{\text{total}} = (1.00 \text{ atm}) + (3.861988 \text{ atm}) = 4.861988 \text{ atm (unrounded)}$
 Use Boyle's Law to find the volume change of the diver's lungs:
 $V_f = V_i P_i / P_f = V_i (4.861988 \text{ atm}) / (1.00 \text{ atm}) = 4.861988 = \mathbf{4.86 V_i}$
 To find the depth to which the diver could ascend safely, use the given safe expansion factor (1.5) and the pressure at 125 ft, P_{125} , to find the safest ascended pressure, P_{safe} .
 $P_{125} / P_{\text{safe}} = 1.5$
 $P_{\text{safe}} = P_{125} / 1.5 = (4.861988 \text{ atm}) / 1.5 = 3.241325 \text{ atm (unrounded)}$
 Convert the pressure in atm to pressure in ft of seawater using the conversion factors above. Subtract this distance from the initial depth to find how far the diver could ascend.
 $P(\text{H}_2\text{O}) = [(4.861988 - 3.241325) \text{ atm}] \times (760 \text{ mmHg} / 1 \text{ atm}) \times [(13.5 \text{ g Hg/mL}) / (1.04 \text{ g H}_2\text{O/mL})]$
 $\times (10^{-3} \text{ m} / 1 \text{ mm}) \times (1.094 \text{ yd} / 1 \text{ m}) \times (3 \text{ ft} / 1 \text{ yd}) = 52.4741 \text{ ft (unrounded)}$
 Therefore, the diver can safely ascend 52.5 ft to a depth of $(125 - 52.4741) = 72.5259 = \mathbf{73 \text{ ft}}$.

- 5.109 First write the balanced equation: $2 \text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
 According to the description in the problem, a given volume of peroxide solution (0.100 L) will release a certain number of "volumes of oxygen gas" (20). Assume that 20 is exact.
 0.100 L solution will produce $(20 \times 0.100 \text{ L}) = 2.00 \text{ L O}_2 \text{ gas}$
 You can convert volume of $\text{O}_2(\text{g})$ to moles of gas using the ideal gas equation.

$$\text{Moles O}_2 = PV / RT = \frac{(1.00 \text{ atm})(2.00 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(273 \text{ K})} = 8.92327 \times 10^{-2} \text{ mol O}_2 \text{ (unrounded)}$$

 The grams of hydrogen peroxide can now be solved from the stoichiometry using the balanced chemical equation.
 $\text{Mass H}_2\text{O}_2 = (8.92327 \times 10^{-2} \text{ mol O}_2) (2 \text{ mol H}_2\text{O}_2 / 1 \text{ mol O}_2) (34.02 \text{ g H}_2\text{O}_2 / 1 \text{ mol H}_2\text{O}_2)$
 $= 6.071395 = \mathbf{6.07 \text{ g H}_2\text{O}_2}$

- 5.121 V and T are not given, so the ideal gas equation cannot be used. The total pressure of the mixture is given:
 $P_{\text{total}} = P_{\text{krypton}} + P_{\text{carbon dioxide}} = 0.708 \text{ atm}$
 The NaOH absorbed the CO_2 leaving the Kr, thus $P_{\text{krypton}} = 0.250 \text{ atm}$
 $P_{\text{carbon dioxide}} = P_{\text{total}} - P_{\text{krypton}} = 0.708 \text{ atm} - 0.250 \text{ atm} = 0.458 \text{ atm}$
 Determining mole fractions:
 Carbon dioxide: $X = (0.458 \text{ atm}) / (0.708 \text{ atm}) = 0.64689 \text{ (unrounded)}$
 Krypton: $X = (0.250 \text{ atm}) / (0.708 \text{ atm}) = 0.353107 \text{ (unrounded)}$
 Relative mass fraction = $[(0.353107) (83.80 \text{ g Kr/mol})] / [(0.64689) (44.01 \text{ g CO}_2 / \text{mol})]$
 $= 1.039366 \text{ (unrounded)}$
 $35.0 \text{ g} = x \text{ g CO}_2 + (1.039366 x) \text{ g Kr}$
 $\text{Grams CO}_2 = (35.0 \text{ g}) / (2.039366) = 17.16219581 = \mathbf{17.2 \text{ g CO}_2}$
 $\text{Grams Kr} = 35.0 \text{ g} - 17.162 \text{ g CO}_2 = 17.83780419 = \mathbf{17.8 \text{ g Kr}}$

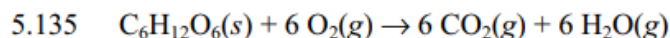
- 5.126 The root mean speed is related to temperature and molar mass as shown in equation:

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$u_{\text{rms Ne}} = \sqrt{\frac{3(8.314 \text{ J/mol} \cdot \text{K})(370 \text{ K})}{(20.18 \text{ g/mol})} \left(\frac{10^3 \text{ g}}{\text{kg}}\right) \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}}\right)} = 676.24788 = \mathbf{676 \text{ m/s Ne}}$$

$$u_{\text{rms Ar}} = \sqrt{\frac{3(8.314 \text{ J/mol} \cdot \text{K})(370 \text{ K})}{(39.95 \text{ g/mol})} \left(\frac{10^3 \text{ g}}{\text{kg}}\right) \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}}\right)} = 480.6269 = \mathbf{481 \text{ m/s Ar}}$$

$$u_{\text{rms He}} = \sqrt{\frac{3(8.314 \text{ J/mol} \cdot \text{K})(370 \text{ K})}{(4.003 \text{ g/mol})} \left(\frac{10^3 \text{ g}}{\text{kg}}\right) \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}}\right)} = 1518.356 = \mathbf{1.52 \times 10^3 \text{ m/s He}}$$



a) First, find the number of moles of carbon dioxide and use the T and P given to calculate volume from the ideal gas equation.

V =

$$\frac{\left[18.0 \text{ g Glucose} \left(\frac{1 \text{ mol Glucose}}{180.16 \text{ g Glucose}} \right) \left(\frac{6 \text{ mol CO}_2}{1 \text{ mol Glucose}} \right) \right] \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) ((273 + 35) \text{ K})}{(780. \text{ torr})} \left(\frac{760 \text{ torr}}{1 \text{ atm}} \right)$$

$$= 14.76992 = \mathbf{14.8 \text{ Liters CO}_2}$$

This solution assumes that partial pressure of O_2 does not interfere with the reaction conditions.

b) From the stoichiometric ratios, calculate the moles of each gas and then use Dalton's law of partial pressures to determine the pressure of each gas.

Moles $\text{C}_6\text{H}_{12}\text{O}_6 = (9.0 \text{ g glucose}) (1 \text{ mol glucose} / 180.16 \text{ g glucose}) = 0.049956 \text{ mol glucose (unrounded)}$

Moles of other substances = $(0.049956 \text{ mol glucose}) (6 \text{ mol} / 1 \text{ mol glucose}) = 0.299736 \text{ mol (unrounded)}$

At 35°C , the vapor pressure of water is 42.2 torr. No matter how much water is produced, the partial pressure of H_2O will still be 42.2 torr. The remaining pressure, $780 \text{ torr} - 42.2 \text{ torr} = 737.8 \text{ torr (unrounded)}$ is the sum of partial pressures for O_2 and CO_2 . Since the mole fractions of O_2 and CO_2 are equal, their pressures must be equal, and be one-half of sum of the partial pressures just found.

$P_{\text{water}} = \mathbf{42.2 \text{ torr}}$

$(737.8 \text{ torr}) / 2 = 368.9 = \mathbf{3.7 \times 10^2 \text{ torr } P_{\text{oxygen}} = P_{\text{carbon dioxide}}}$

5.140 To find the number of steps through the membrane, calculate the molar masses to find the ratio of effusion rates. This ratio is the enrichment factor for each step.

From the margin note, $M_{235} = 349.03 \text{ g/mol}$ and $M_{238} = 352.04 \text{ g/mol}$

$$\text{Rate } ^{235}\text{U} / \text{Rate } ^{238}\text{U} = \sqrt{\frac{\text{Molar Mass } ^{238}\text{UF}_6}{\text{Molar Mass } ^{235}\text{UF}_6}} = \sqrt{\frac{352.04 \text{ g/mol}}{349.03 \text{ g/mol}}}$$

$$= 1.004302694 \text{ enrichment factor (unrounded)}$$

Therefore, the abundance of $^{235}\text{UF}_6$ after one membrane is $0.72\% \times 1.004302694$, and after "N" membranes:

Abundance of $^{235}\text{UF}_6$ after "N" membranes = $0.72\% * (1.004302694)^N$

Desired abundance of $^{235}\text{UF}_6 = 3.0\% = 0.72\% * (1.004302694)^N$

Solving for N:

$$4.16667 = (1.004302694)^N$$

$$\ln 4.16667 = \ln (1.004302694)^N$$

$$\ln 4.16667 = N * \ln (1.004302694)$$

$$N = (\ln 4.16667) / (\ln 1.004302694) = 332.392957 = \mathbf{332 \text{ steps}}$$

5.142 The amount of each gas that leaks from the balloon is proportional to its effusion rate. Using 45% or 0.45 as the rate for H_2 , the rate for O_2 can be determined from Graham's Law.

$$\text{Rate oxygen} / \text{Rate hydrogen} = \sqrt{\frac{\text{Molar Mass } \text{H}_2}{\text{Molar Mass } \text{O}_2}} = \sqrt{\frac{2.016 \text{ g/mol}}{32.00 \text{ g/mol}}} = \frac{\text{Rate } \text{O}_2}{\left(\frac{45\%}{100\%} \right)}$$

$$\text{Rate } \text{O}_2 = 0.112949(100\%) = 11.2949 \text{ (unrounded)}$$

$$\text{O}_2 / \text{H}_2 = (100 - 11.2949) / (100 - 45) = 1.6128 = \mathbf{1.6}$$

5.146 The balanced chemical equation is: $7 \text{F}_2(\text{g}) + \text{I}_2(\text{s}) \rightarrow 2 \text{IF}_7(\text{g})$

It will be necessary to determine the limiting reactant.

$$\begin{aligned} \text{Moles IF}_7 \text{ from F}_2 &= \frac{(350. \text{ torr})(2.50 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(250. \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) \left(\frac{2 \text{ mol IF}_7}{7 \text{ mol F}_2}\right) \\ &= 0.016026668 \text{ mol IF}_7 \text{ (unrounded)} \end{aligned}$$

$$\begin{aligned} \text{Moles IF}_7 \text{ from I}_2 &= (2.50 \text{ g I}_2) (1 \text{ mol I}_2 / 253.8 \text{ g I}_2) (2 \text{ mol IF}_7 / 1 \text{ mol I}_2) \\ &= 0.019700551 \text{ mol IF}_7 \text{ (unrounded)} \end{aligned}$$

F_2 is limiting.

$$\begin{aligned} \text{Mole I}_2 \text{ remaining} &= (2.50 \text{ g I}_2) (1 \text{ mol I}_2 / 253.8 \text{ g I}_2) - \frac{(350. \text{ torr})(2.50 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(250. \text{ K})} \left(\frac{1 \text{ atm}}{760 \text{ torr}}\right) \left(\frac{1 \text{ mol I}_2}{7 \text{ mol F}_2}\right) \\ &= 1.83694 \times 10^{-3} \text{ mol I}_2 \text{ (unrounded)} \end{aligned}$$

$$\begin{aligned} \text{Total moles of gas} &= (0 \text{ mol F}_2) + (0.016026668 \text{ mol IF}_7) + (1.83694 \times 10^{-3} \text{ mol I}_2) \\ &= 0.0178636 \text{ mol gas (unrounded)} \end{aligned}$$

$$P_{\text{total}} = nRT / V = \frac{(0.0178636 \text{ mol}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) (550. \text{ K})}{(2.50 \text{ L})} = 0.322652 = \mathbf{0.323 \text{ atm}}$$

$$P_{\text{iodine}} = X_{\text{iodine}} P_{\text{total}} = [(1.83694 \times 10^{-3} \text{ mol I}_2) / (0.0178636 \text{ mol})] (0.322652 \text{ atm}) = 0.03317877 = \mathbf{0.0332 \text{ atm}}$$